

PAUL MITCHELL

AI SYSTEMS ENGINEER · DISTRIBUTED SYSTEMS · SECURE AI INFRASTRUCTURE

Riverdale, MD · (413) 204-0493 · pjmitchell@wm.edu · github.com/pjmitchell7 · U.S. Citizen | Eligible for Security Clearance

PROFESSIONAL SUMMARY

William & Mary M.S. in Computer Science candidate (3.95 GPA, AI/ML & Systems) and clearable U.S. citizen targeting enterprise AI Systems Engineering, Defense Technology & Solutions practices. Combines deep hands-on Linux systems administration (100+ instructional servers, Prometheus/Grafana, Puppet) with production machine learning pipelines, pre-inference LLM security proxies, and cloud microservices.

EDUCATION

William & Mary — M.S. in Computer Science (AI/ML & Systems)

Expected Dec 2026 · GPA: 3.95

William & Mary — B.S. in Computer Science

Graduated Dec 2024

TECHNICAL & SYSTEMS EXPERIENCE

IT Administrator — William & Mary Technology Support Center

Feb 2025 – Present | Williamsburg, VA

- Administer 100+ production Linux instructional servers; manage configuration automation, networking, DNS, and user isolation via Puppet.
- Deployed Prometheus and Grafana telemetry dashboards monitoring cluster compute load, memory pressure, and node health, accelerating incident triage.
- Engineered automated system verification and health probe scripts resolving recurring hardware and network route bottlenecks.

Data Analytics & AI Extern — Extern (Amazon-Sponsored Project)

Jun 2025 – Sep 2025 | Remote

- Engineered NLP topic discovery pipelines combining fine-tuned DistilBERT (F1 = 0.89) and BERTopic clustering on ~20,000 unstructured text records.
- Trained predictive classification models using LightGBM and deployed an interactive Streamlit telemetry application for leadership analytics.
- Implemented data validation pipelines on AWS cloud storage (S3) ensuring consistent feature representation across high-volume batches.

KEY ENGINEERING PROJECTS

ShadowPrompt: Pre-Inference LLM Security & Deceptive Defense Proxy

2026 · pjmitchell7.github.io/shadowprompt

- Architected a pre-inference defense proxy inspecting raw Unicode streams in memory at < 0.1ms P99 latency, neutralizing zero-width steganography, directional overrides (RLO), and chat template breakouts before tokenization.
- Integrated deceptive honeypot sandboxes seeded with canary credentials to waste automated adversary compute without alerting attackers.
- Generated compliant STIX 2.1 threat intelligence bundles and ArcSight CEF syslog feeds for direct ingestion into Splunk and Microsoft Sentinel SIEMs.

PromptShield: Tokenization-Aware LLM Security Middleware

2026 · Python, FastAPI, scikit-learn

- Designed an asynchronous FastAPI middleware layer intercepting prompt injections with 0.27–1.11ms evaluation latency.
- Benchmarked false-positive resistance against complex SQL, multi-language dialogue, and enterprise code samples.

TECHNICAL SKILLS & INFRASTRUCTURE

Systems & Infrastructure: Linux Systems Administration (100+ nodes), Prometheus, Grafana, Puppet, Bash, Docker, Networking, DNS, Git.

AI & Machine Learning: Python, PyTorch, Hugging Face, DistilBERT, Sentence-BERT, BERTopic, LightGBM, RAG Architectures, LangGraph, vLLM.

Backend & Cloud: FastAPI, REST Microservices, AWS (S3, EC2, Bedrock), SQL (PostgreSQL), Redis, STIX 2.1, ArcSight CEF.